

Electromagnetism 2

(spring semester 2026)

Lecture 1

Static electric and magnetic fields

- ❖ Electrostatic field, electrostatic potential, Gauss law
- ❖ Static magnetic field, Ampere's law
- ❖ Lorentz force and relativity

Practical details

Course materials:

- ❖ The course covers three main topics.
 - ✓ **Lectures 1–6:** revision and extension of EM1, leading to **Maxwell's equations in free space**.
 - ✓ **Lectures 7–12:** conductors, dielectric and magnetic media, leading to **Maxwell's equations in media**.
 - ✓ **Remaining lectures:** **electromagnetic waves** & their properties.
- ❖ Materials available on Canvas: lecture notes; five non-assessed problem sheets; past examination papers.

Links to other modules:

- ❖ Knowledge of EM1, vector calculus, geometry (e.g. solid angles) is assumed.
- ❖ EM2 is essential for many optional Y3/Y4 modules: radiation and relativity; images and communication, physical principles of radar, ...

Reading list (ranked from basic to advanced):

- ❖ I.S. Grant, W.R. Phillips: Electromagnetism, 2nd ed. (Wiley)
- ❖ R. Feynman, R. Leighton, M. Sands: The Feynman Lectures on Physics (Vol II)
- ❖ W.N. Cottingham, D.A. Greenwood: Electricity and Magnetism (Cambridge)
- ❖ D.J. Griffiths: Introduction to Electrodynamics, 4th ed. (Cambridge)
- ❖ J.D. Jackson: Classical Electrodynamics, 3rd ed. (Wiley)

The system of units

The system of units

- ❖ The SI system is used in this course.
- ❖ Some textbooks use the Gaussian system.
- ❖ The SI is convenient for practical computations but has fundamental drawbacks wrt the Gaussian system.
 - ✓ Conversion factors are introduced:
the *electric constant* $\epsilon_0 \approx 8.85 \times 10^{-12} \text{ C/(Vm)}$ and
the *magnetic constant* $\mu_0 \approx 4\pi \times 10^{-7} \text{ H/m}$.
 - ✓ Only their product has a physical meaning ($\epsilon_0 \mu_0 = 1/c^2$).
 - ✓ Different units used for electric (**E**, **D**) and magnetic (**B**, **H**) fields (**V/m**, **C/m²**, **T**, **A/m**), though the distinction between electric vs magnetic fields is relative.

Electrostatic field (1)

Coulomb's law (1785): force acting on a point charge q in the field of a point charge Q .

$$\vec{F} = \frac{Qq}{4\pi\epsilon_0} \cdot \frac{\vec{r}}{r^3}$$

Very strong wrt gravity! Two protons: $F_G/F_C \sim 10^{-36}$.

Electrostatic field of a point charge Q :
[unit: V/m]

$$\vec{E}(\vec{r}) = \frac{\vec{F}}{q} = \frac{Q}{4\pi\epsilon_0} \cdot \frac{\vec{r}}{r^3}$$

The $\mathbf{E}$ -field, introduced formally above, is a real physical substance. Space is not empty! EM field can carry energy (e.g. as EM waves).

$\vec{E}(\vec{r})$ is a **conservative vector field**: we can define a scalar potential φ ,

$$\vec{E} = - \left(\frac{\partial \varphi}{\partial x} \vec{e}_x + \frac{\partial \varphi}{\partial y} \vec{e}_y + \frac{\partial \varphi}{\partial z} \vec{e}_z \right) = -\text{grad } \varphi = -\nabla \varphi$$

This also means that $\oint \vec{E} d\vec{l} = 0$ for any closed curve.

Physical meaning of φ_A in electrostatics: work against the field required to bring a unit test charge from infinity to the point A .

For a point charge Q , $\varphi(r) = \int_r^\infty \frac{Q}{4\pi\epsilon_0 t^2} dt = -\frac{Q}{4\pi\epsilon_0 t} \Big|_r^\infty = \frac{Q}{4\pi\epsilon_0 r}$ 3

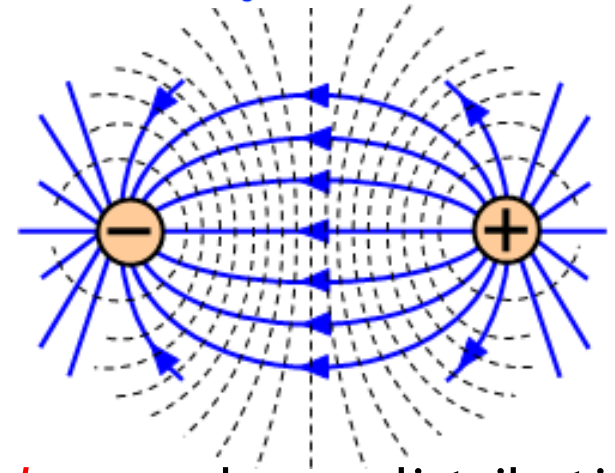
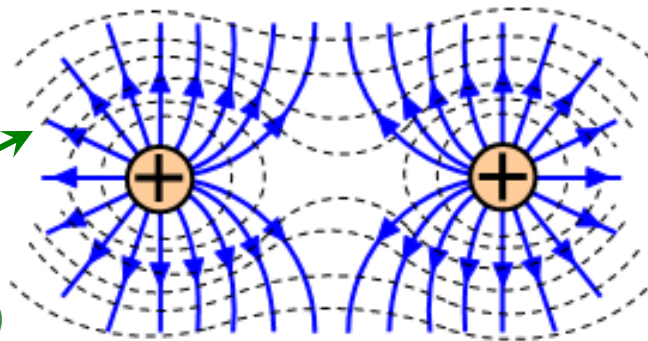
Electrostatic field (2)

The *principle of superposition*: for multiple point charges Q_i ,

the total field is $\vec{E} = \sum_i \vec{E}_i$, therefore $\varphi = \sum_i \varphi_i$

Equipotentials (perpendicular to field lines)

Field lines
(tangent to
the field vector)



Now we can compute electric field for any *known* charge distribution.

Example: a long wire with a constant linear charge density $\lambda > 0$ [C/m].

charged wire

$$E = \frac{\lambda}{4\pi\epsilon_0} \int_{-\infty}^{+\infty} \frac{dx}{l^2} \cos \alpha = \frac{\lambda}{4\pi\epsilon_0} \int_{-\infty}^{+\infty} d(R \tan \alpha) \frac{\cos^2 \alpha}{R^2} \cos \alpha$$

$$= \frac{\lambda}{4\pi\epsilon_0 R} \int_{-\pi/2}^{+\pi/2} \cos \alpha d\alpha = \frac{\lambda}{2\pi\epsilon_0 R}$$

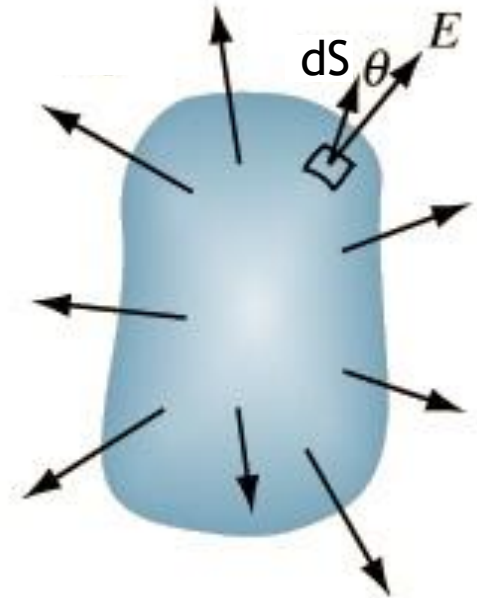
Gauss law

Gauss law valid for the electric field
(universally valid, also for non-static fields):

$$\underbrace{\int_S \vec{E} d\vec{S}}_{\text{Electric flux through any closed surface}} = \frac{1}{\epsilon_0} \underbrace{\int_V \rho dV}_{\text{Total electric charge within the surface}}$$

Electric flux through
any closed surface

Total electric charge
within the surface

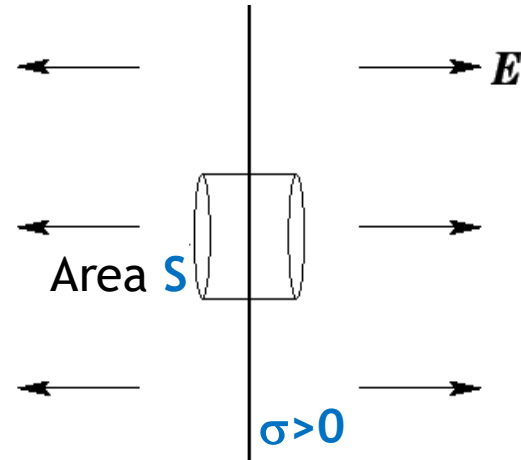


- ❖ Gauss law is *equivalent to Coulomb's law*.
- ❖ Gauss law simplifies field computations for systems with a sufficient degree of symmetry.
- ❖ Gauss law is not sufficient for computations in general case: a single scalar relation for the three unknowns E_x , E_y , E_z .
- ❖ Similarity of Coulomb's law and Newton's law of gravity ($F \sim 1/r^2$): Gauss law applies also to the gravitational field.

Gauss law: examples

- 1) **Infinite plane**, surface charge density σ [C/m²].
Symmetry: field is perpendicular to the plate.

Gauss law: $2SE = \sigma S/\epsilon_0$; $E = \frac{\sigma}{2\epsilon_0}$



- 2) For a **planar capacitor**, use superposition:

$E = \frac{\sigma}{\epsilon_0}$ between the plates; $E = 0$ outside.

- 3) **Uniformly charged solid sphere** (radius R , charge Q).

Inside the sphere,

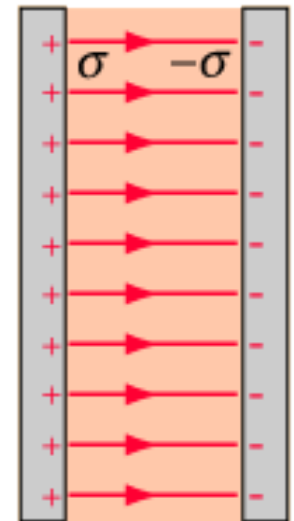
$$4\pi r^2 E = \frac{Q}{\epsilon_0} \left(\frac{r}{R}\right)^3$$

$$E(r) = \frac{Qr}{4\pi\epsilon_0 R^3}$$

Outside the sphere,

$$4\pi r^2 E = \frac{Q}{\epsilon_0}$$

$$E(r) = \frac{Q}{4\pi\epsilon_0 r^2}$$



Static magnetic field

The *current density* vector: $\vec{j} = \rho \vec{v}$ [A/m²]

Right-hand side: product of charge density [C/m³] and velocity [m/s].

Biot-Savart law (1820): magnetic field due to a *linear elementary current*, or a *volume elementary current*:

$$\vec{B}(\vec{r}) = \frac{\mu_0}{4\pi} \cdot \frac{Id\vec{l} \times \vec{r}}{r^3} = \frac{\mu_0}{4\pi} \cdot \frac{(\vec{j} \times \vec{r})dV}{r^3} \quad [\text{unit: T}]$$

Magnetic force acting on an elementary current:

$$\vec{F} = Id\vec{l} \times \vec{B} = (\vec{j} \times \vec{B})dV$$

Equivalently, *Lorentz force* acting on a charge q moving in a magnetic field with velocity $\vec{v}$: $\vec{F} = q\vec{v} \times \vec{B}$

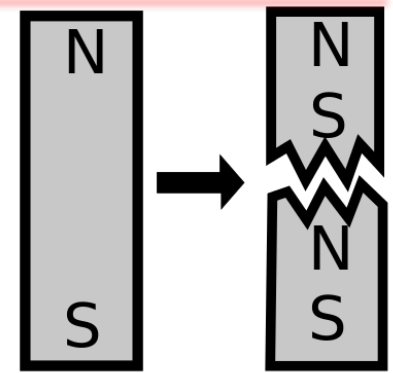
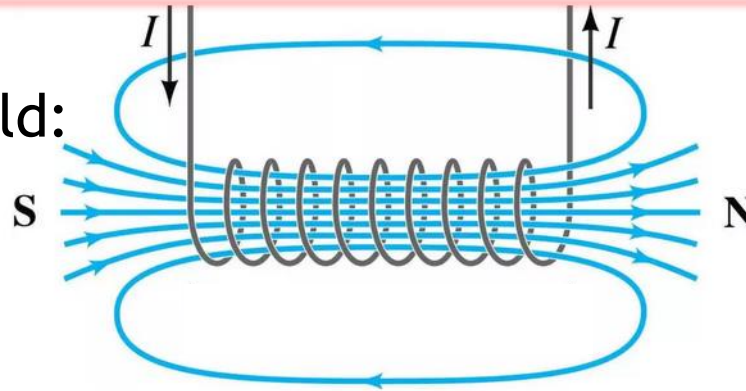
Principle of superposition: $\vec{B} = \sum_i \vec{B}_i$

Example: attraction (repulsion) of parallel (antiparallel) currents. 7

Magnetic field lines; Ampere's law

Continuous field lines,
no sources of the **B** field:

$$\oint_S \vec{B} d\vec{S} = 0$$

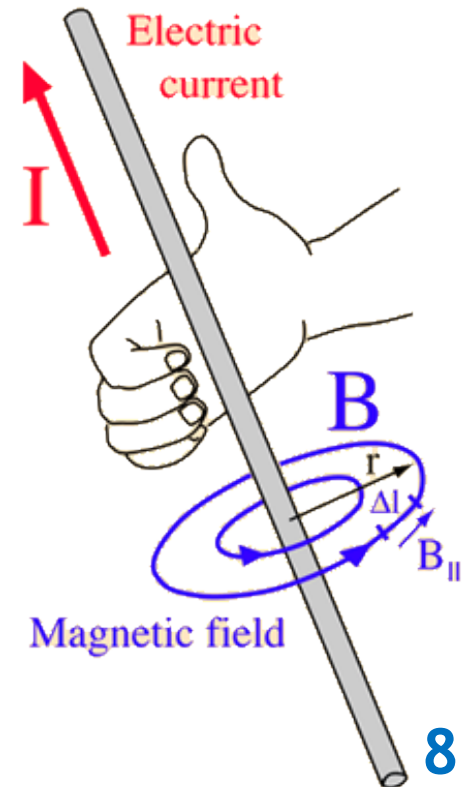


Ampere's law for the magnetic field
produced by steady currents:

$$\oint_L \vec{B} d\vec{l} = \mu_0 \oint_S \vec{j} d\vec{S}$$

Line integral of
the magnetic field
around a closed curve

Current through
any surface enclosed
by the curve



Sign convention: the direction of **B** relative to **j**
is determined by the *right-hand rule*.

Ampere's law: examples

1) Straight infinite wire: cylindrical symmetry.

Outside the wire,

$$2\pi r B = \mu_0 I$$

$$B = \frac{\mu_0 I}{2\pi r}$$

For $I = 1 \text{ A}$ and $r = 2 \text{ cm}$,

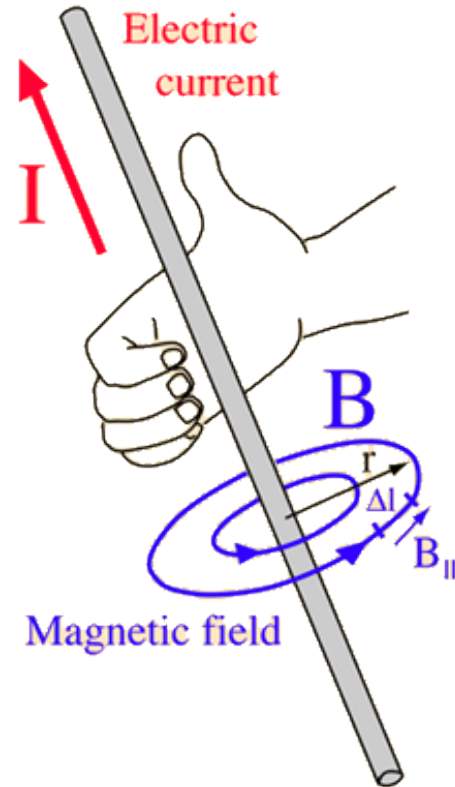
$$B = 4\pi \times 10^{-7} \text{ H}\cdot\text{m}^{-1} \cdot 1 \text{ A} / (2\pi \cdot 0.02 \text{ m}) = 10^{-5} \text{ T}$$

(comparable to the Earth's magnetic field)

Inside the wire, assuming uniform current density $\mathbf{j}$,

$$2\pi r B = \mu_0 I (r/R)^2$$

$$B = \frac{\mu_0 I r}{2\pi R^2} = \frac{\mu_0 j r}{2}$$

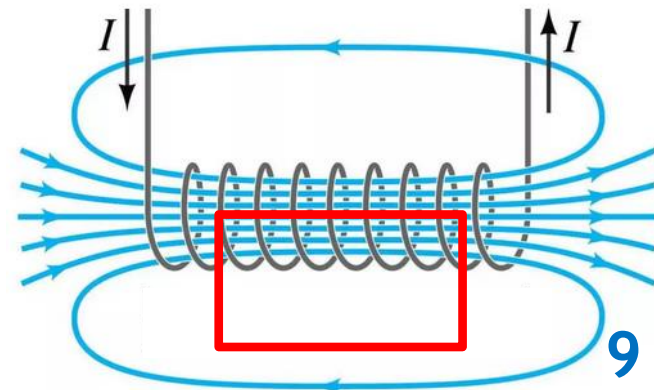


2) Long solenoid with N turns per unit length. Outside, $B=0$ by considering it part of a very large toroid and using symmetry.

Inside the solenoid,

$$BL = \mu_0 \cdot NL \cdot I$$

$$B = \mu_0 NI$$



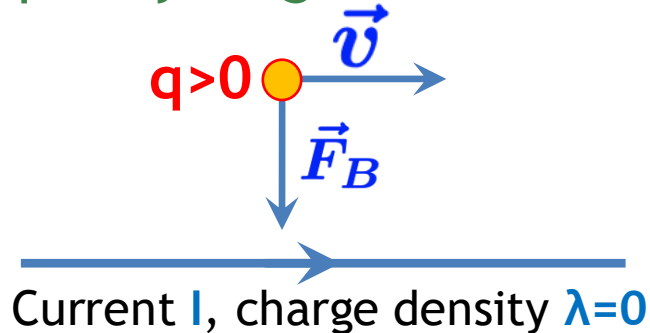
Lorentz force and relativity

- ❖ Lorentz force acting on a point charge q in $\mathbf{E}$ and $\mathbf{B}$ fields:

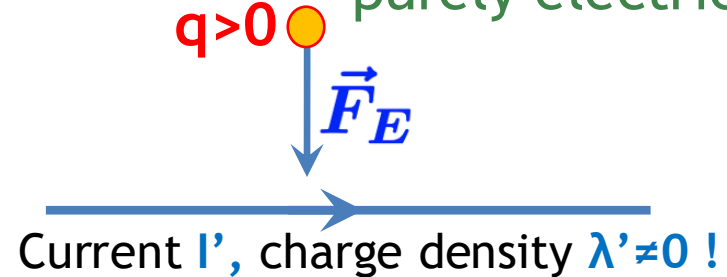
$$\vec{F} = \vec{F}_E + \vec{F}_B = q(\vec{E} + \vec{v} \times \vec{B})$$

- ❖ Force on a charge moving along a wire carrying a current:

Lab reference frame S :
purely magnetic force



Reference frame of the charge S' :
 $v=0$, $F_B=0$, therefore the force is
purely electric



- ❖ The wire becomes charged ($\lambda' \neq 0$) in the reference frame S' !
This is due to *relativistic length contraction*.
- ❖ Electric and magnetic forces are two manifestations of the same phenomenon, *electromagnetism*.
- ❖ Electromagnetism is intrinsically *relativistic*.

Summary

- ❖ Laws of electrostatics and magnetostatics *in free space*, *in the integral form*, useful for practical computations:

Gauss law (*universally valid*)

$$\oint_S \vec{E} d\vec{S} = \frac{1}{\epsilon_0} \int_V \rho dV$$

Conservative nature of the **E**-field (*static field only*)

$$\oint_L \vec{E} d\vec{l} = 0$$

Absence of magnetic poles (*universally valid*)

$$\oint_S \vec{B} d\vec{S} = 0$$

Ampere's law (*static field only*)

$$\oint_L \vec{B} d\vec{l} = \mu_0 \int_S \vec{j} d\vec{S}$$